

## **Paper 1**

### **Title :**

## **Hybrid Multi-Modal Skin Cancer Classification using CNN, Vision Transformer and Explainable AI**

### **Abstract**

Skin cancer is one of the most common and rapidly increasing forms of cancer worldwide, making early and accurate diagnosis essential for improving patient survival. Existing deep learning approaches primarily rely on either Convolutional Neural Networks (CNNs) or Vision Transformers (ViTs), which limits their ability to simultaneously capture detailed local features and global contextual information. Moreover, many existing systems classify skin lesions using only dermoscopic images while ignoring valuable clinical metadata such as patient age, gender, and lesion location. In addition, the lack of explainability in AI models reduces clinicians' trust in automated diagnosis. This project proposes a Hybrid Multi-Modal Skin Cancer Classification Framework that combines CNN and Vision Transformer (ViT) architectures to learn complementary local and global image features. The proposed system integrates dermoscopic images with patient metadata through a multi-modal feature fusion strategy to improve classification accuracy across multiple skin cancer categories. To enhance transparency and clinical reliability, Explainable Artificial Intelligence (XAI) techniques such as Grad-CAM and LIME are incorporated to highlight the important image regions and influential factors responsible for the model's predictions. The system also generates confidence scores, risk levels, and a doctor-friendly diagnostic report through a real-time Streamlit web application. The proposed framework aims to provide an accurate, interpretable, and clinically reliable computer-aided diagnostic system that supports dermatologists in the early detection and classification of skin cancer while improving decision-making and patient outcomes.

### **Introduction**

Skin cancer is one of the most common and rapidly increasing types of cancer worldwide. It develops when abnormal skin cells grow uncontrollably due to factors such as excessive exposure to ultraviolet (UV) radiation, genetic predisposition, and environmental influences. Among the different types of skin cancer, melanoma is the most dangerous because it can spread quickly to other parts of the body if not detected at an early stage. Therefore, early and accurate diagnosis is essential for improving patient survival and reducing treatment costs.

In recent years, Artificial Intelligence (AI) and Deep Learning have become powerful tools for automated skin cancer diagnosis. Convolutional Neural Networks (CNNs) have demonstrated excellent performance in extracting local image features such as color, texture, and lesion boundaries. However, CNNs have limited capability in capturing long-range relationships and global contextual information within skin lesion images. Vision Transformers (ViTs), on the other hand, utilize self-attention mechanisms to learn global image representations but may overlook fine-grained local details when used independently.

To overcome these limitations, this project proposes a Hybrid Multi-Modal Skin Cancer Classification Framework that combines the strengths of CNN and Vision Transformer. The CNN extracts detailed local features, while the Vision Transformer captures global contextual information. The extracted features are fused to generate a more comprehensive representation of skin lesions, resulting in improved classification performance.

In addition to dermoscopic images, the proposed system incorporates patient clinical metadata such as age, gender, and lesion location through a multi-modal learning approach. Combining image features with patient information enables the model to make more accurate and clinically relevant predictions than image-only methods.

To increase the transparency and reliability of the proposed system, Explainable Artificial Intelligence (XAI) techniques such as Grad-CAM and LIME are integrated. These methods visualize the important regions of the skin lesion and explain the factors influencing the model's prediction, thereby helping dermatologists understand and trust the system's decisions.

Finally, the proposed model is deployed as a user-friendly Streamlit web application that allows users to upload skin lesion images and receive disease predictions along with confidence scores, risk levels, visual explanations, and a doctor-friendly diagnostic report. The overall objective is to develop an accurate, interpretable, and practical computer-aided diagnostic system that supports dermatologists in the early detection and classification of skin cancer.